

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn. preprocessing import StandardScaler
```

```
from google.colab import files
uploaded = files.upload()
```

[Choose files](#) clv_data.csv
clv_data.csv(text/csv) - 194373 bytes, last modified: 30/06/2026 - 100% done
Saving clv_data.csv to clv_data (1).csv

```
import pandas as pd
import io
df = pd.read_csv("clv_data.csv")

df.head()
```

	Unnamed: 0	id	age	gender	income	days_on_platform	city	purchases	
0	0	0	NaN	Male	126895	14.0	San Francisco	0	
1	1	1	NaN	Male	161474	14.0	Tokyo	0	
2	2	2	24.0	Male	104723	34.0	London	1	
3	3	3	29.0	Male	43791	28.0	London	2	
4	4	4	18.0	Female	132181	26.0	London	2	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.shape
```

(5000, 8)

```
df.columns
```

Index(['Unnamed: 0', 'id', 'age', 'gender', 'income', 'days_on_platform',
 'city', 'purchases'],
 dtype='object')

```
df.dtypes
```

	0
Unnamed: 0	int64
id	int64
age	float64
gender	object
income	int64
days_on_platform	float64
city	object
purchases	int64

dtype: object

```
df.describe()
```

	Unnamed: 0	id	age	income	days_on_platform	purchases	
count	5000.000000	5000.000000	2554.000000	5000.000000	4859.000000	5000.000000	
mean	2499.500000	2499.500000	30.202036	79593.336200	24.389998	1.101000	
std	1443.520003	1443.520003	12.129439	60304.056095	18.153388	1.181473	
min	0.000000	0.000000	10.000000	4.000000	1.000000	0.000000	
25%	1249.750000	1249.750000	19.000000	32752.500000	10.000000	0.000000	
50%	2499.500000	2499.500000	30.000000	65989.000000	21.000000	1.000000	
75%	3749.250000	3749.250000	41.000000	115406.500000	35.000000	2.000000	
max	4999.000000	4999.000000	50.000000	388572.000000	111.000000	6.000000	

df.isnull().sum()

	0
Unnamed: 0	0
id	0
age	2446
gender	0
income	0
days_on_platform	141
city	0
purchases	0

dtype: int64

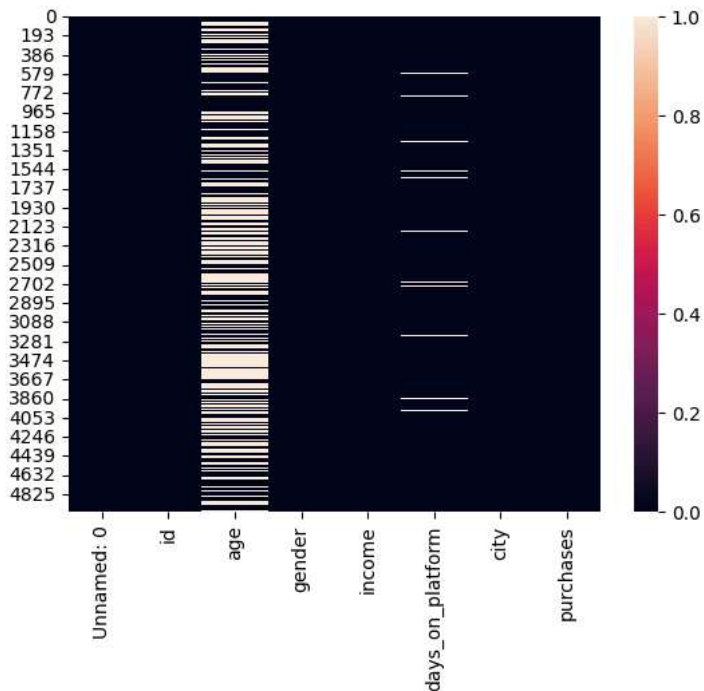
sns.heatmap(df.isnull())
plt.show

```
matplotlib.pyplot.show
def show(*args, **kwargs) -> None
```

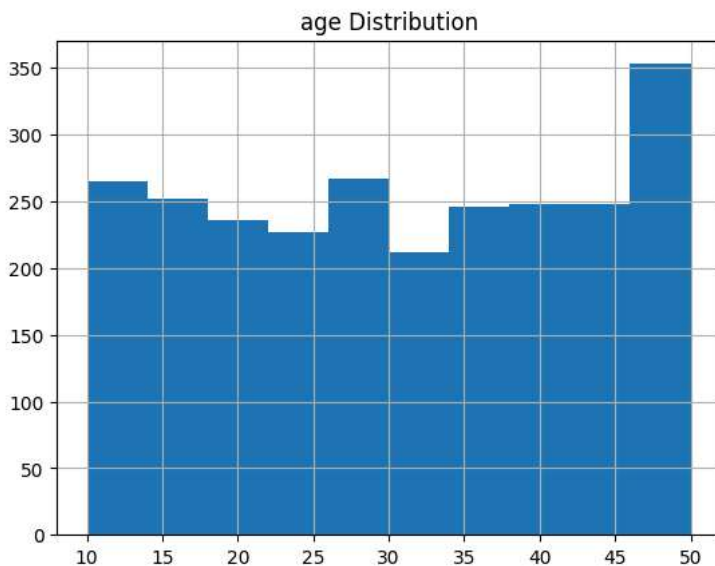
</usr/local/lib/python3.12/dist-packages/matplotlib/pyplot.py>
Display all open figures.

Parameters

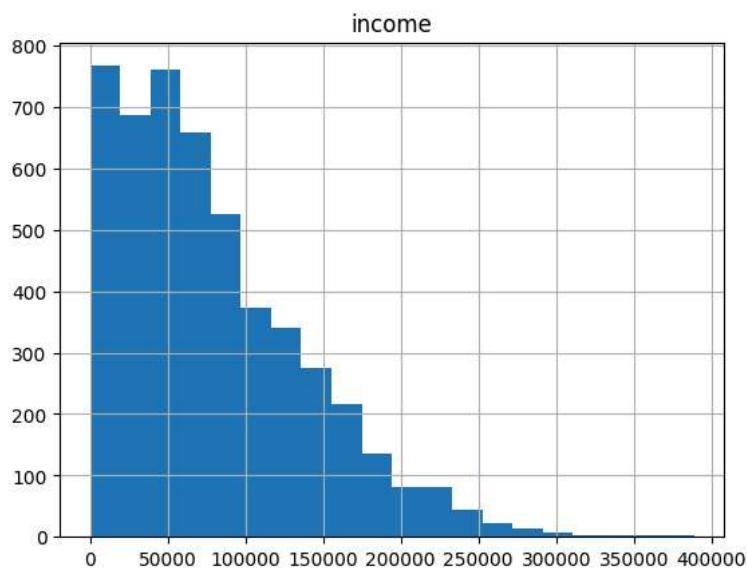
block : bool, optional



```
df["age"].hist(bins=10)
plt.title("age Distribution")
plt.show()
```



```
df["income"].hist(bins=20)
plt.title("income")
plt.show()
```



```
df["income"].var()
```

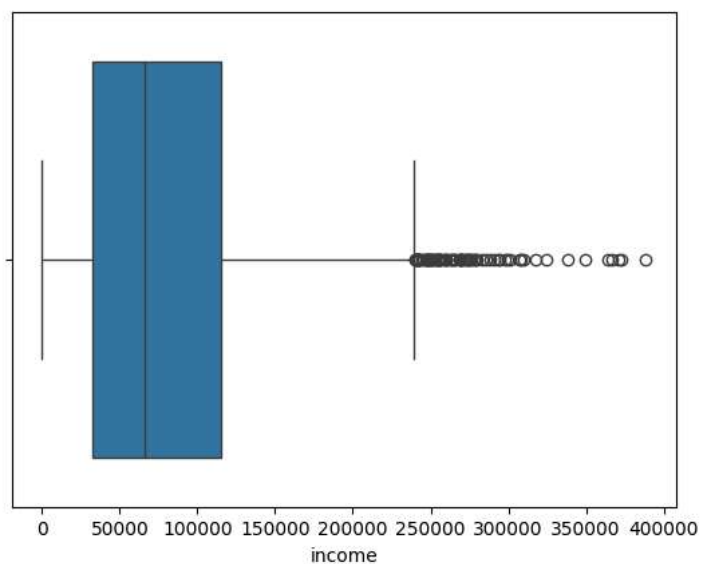
```
3636579181.4510627
```

```
df["income"].std()
```

```
60304.0560945204
```

Start coding or [generate](#) with AI.

```
sns.boxplot(x=df["income"])
plt.show()
```



```
Q1 = df["income"].quantile(0.25)
Q3 = df["income"].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
```

```
lower, upper
```

```
(np.float64(-91228.5), np.float64(239387.5))
```

```
outliers = df[
(df["income"] < lower) |
(df["income"] > upper)
]
outliers
```

Unnamed: 0	id	age	gender	income	days_on_platform	city	purchases	
50	50	50	36.0	Male	255965	22.0	Tokyo	1
96	96	96	NaN	Male	279792	36.0	London	4
200	200	200	46.0	Female	309902	51.0	Tokyo	0
249	249	249	NaN	Female	265599	58.0	London	1
423	423	423	43.0	Female	263840	49.0	London	0
...	...	...	...	...	...	...	...	...
4198	4198	4198	NaN	Female	273579	51.0	New York City	1
4383	4383	4383	11.0	Male	308439	25.0	New York City	0
4723	4723	4723	27.0	Male	264791	14.0	New York City	0
4778	4778	4778	11.0	Male	282909	25.0	Miami	0
4859	4859	4859	16.0	Female	299191	25.0	Miami	2

77 rows × 8 columns

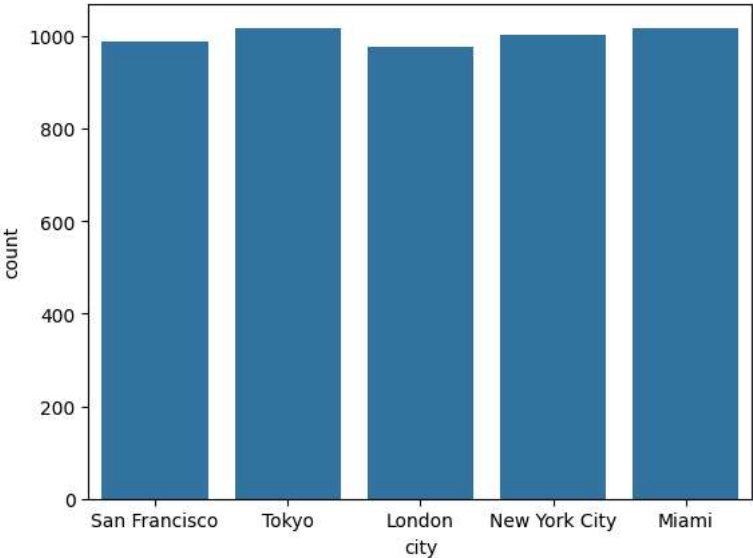
Next steps: [Generate code with outliers](#) [New interactive sheet](#)

```
df_clean = df[
(df["income"] >= lower) &
(df["income"] <= upper)
]
```

```
income = 500000
```

```
df["income"] = np.where(
df["income"] > upper,
upper,
df["income"]
)
```

```
df["gender"].value_counts()
df["city"].value_counts()
sns.countplot(x="city", data=df)
plt.show()
```

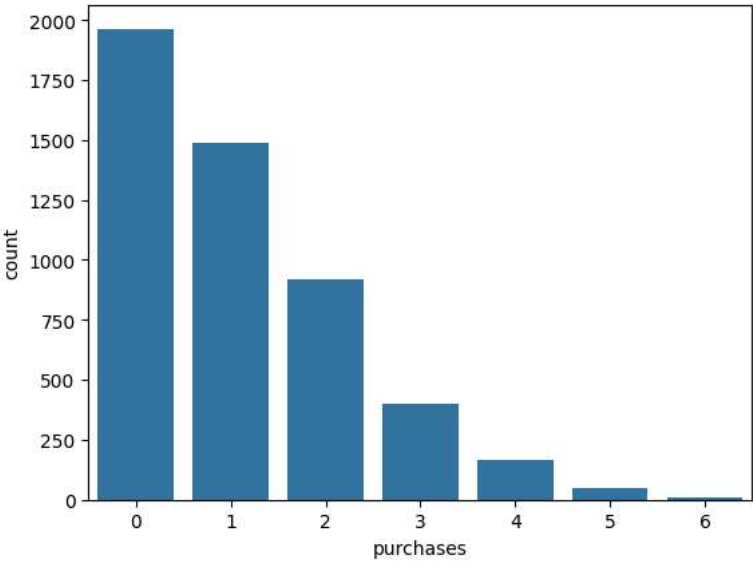


```
df["purchases"].value_counts()
```

count	
purchases	
0	1964
1	1490
2	919
3	400
4	168
5	49
6	10

dtype: int64

```
sns.countplot(x="purchases", data=df)
plt.show()
```



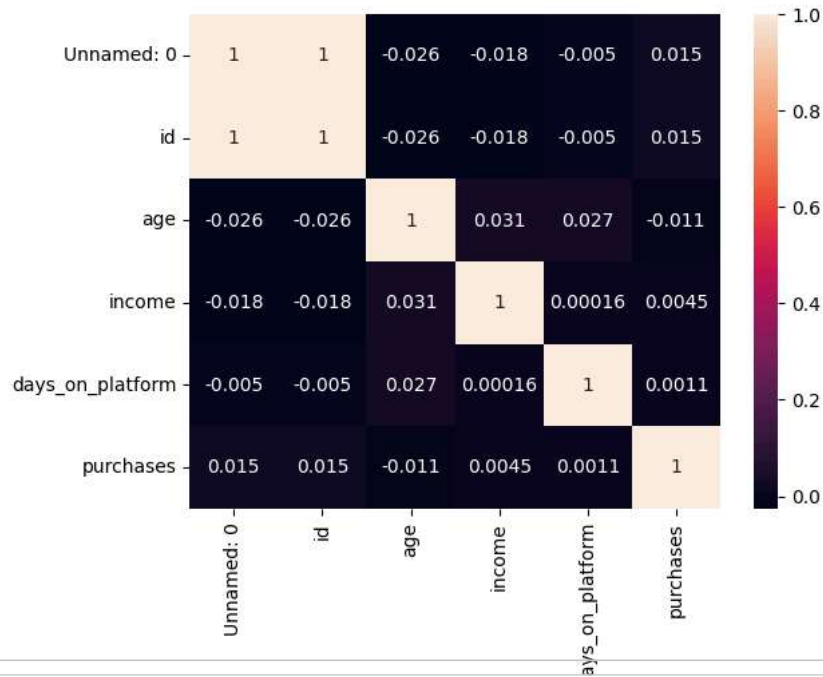
```
from imblearn.over_sampling import SMOTE
```

```
smote = SMOTE()
```

```
df.corr(numeric_only=True)
```

	Unnamed: 0	id	age	income	days_on_platform	purchases	
Unnamed: 0	1.000000	1.000000	-0.025575	-0.018108	-0.004973	0.014985	
id	1.000000	1.000000	-0.025575	-0.018108	-0.004973	0.014985	
age	-0.025575	-0.025575	1.000000	0.030520	0.026916	-0.010595	
income	-0.018108	-0.018108	0.030520	1.000000	0.000164	0.004526	
days_on_platform	-0.004973	-0.004973	0.026916	0.000164	1.000000	0.001107	
purchases	0.014985	0.014985	-0.010595	0.004526	0.001107	1.000000	

```
sns.heatmap(
    df.corr(numeric_only=True),
    annot=True
)
plt.show()
```



```
pd.get_dummies(df, columns=["gender","city"])
```

	Unnamed: 0	id	age	income	days_on_platform	purchases	gender_Female	gender_Male	city_London	city_Miami	city_New York City
0	0	0	NaN	126895.0	14.0	0	False	True	False	False	False
1	1	1	NaN	161474.0	14.0	0	False	True	False	False	False
2	2	2	24.0	104723.0	34.0	1	False	True	True	False	False
3	3	3	29.0	43791.0	28.0	2	False	True	True	False	False
4	4	4	18.0	132181.0	26.0	2	True	False	True	False	False
...	...	...	...	...	...	...	...	...	...	...	...
4995	4995	4995	NaN	212261.0	28.0	1	True	False	False	False	False
4996	4996	4996	NaN	70228.0	12.0	0	False	True	False	False	False
4997	4997	4997	NaN	64995.0	14.0	0	False	True	False	False	True